

# L05: Camera Calibration: DLT

- 3D Point to Pixel: Estimating the Matrix P
  - Corresponding to the 11 intrinsic and extrinsic parameters
- Necessary:  $\geq 6$  points (Each point 2 observation equations - one for each image coordinate)

## Rearrange the DLT Equation

$$\mathbf{x}_i = \underset{3 \times 4}{\mathbf{P}} \mathbf{X}_i = \begin{bmatrix} \boxed{p_{11} \quad p_{12} \quad p_{13} \quad p_{14}} \\ \boxed{p_{21} \quad p_{22} \quad p_{23} \quad p_{24}} \\ \boxed{p_{31} \quad p_{32} \quad p_{33} \quad p_{34}} \end{bmatrix} \mathbf{X}_i$$

Define the three vectors A, B, C as follows

$$\boxed{\mathbf{A}} = \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ p_{14} \end{bmatrix} \quad \boxed{\mathbf{B}} = \begin{bmatrix} p_{21} \\ p_{22} \\ p_{23} \\ p_{24} \end{bmatrix} \quad \boxed{\mathbf{C}} = \begin{bmatrix} p_{31} \\ p_{32} \\ p_{33} \\ p_{34} \end{bmatrix}$$

$$\begin{bmatrix} u_i \\ v_i \\ w_i \end{bmatrix} = \mathbf{x}_i = \mathbf{P} \mathbf{X}_i = \begin{bmatrix} \boxed{\mathbf{A}^\top} \\ \boxed{\mathbf{B}^\top} \\ \boxed{\mathbf{C}^\top} \end{bmatrix} \mathbf{X}_i = \begin{bmatrix} \mathbf{A}^\top \mathbf{X}_i \\ \mathbf{B}^\top \mathbf{X}_i \\ \mathbf{C}^\top \mathbf{X}_i \end{bmatrix}$$

$$\mathbf{x}_i = \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix} = \begin{bmatrix} u_i \\ v_i \\ w_i \end{bmatrix} = \begin{bmatrix} \mathbf{A}^\top \mathbf{X}_i \\ \mathbf{B}^\top \mathbf{X}_i \\ \mathbf{C}^\top \mathbf{X}_i \end{bmatrix}$$



$$x_i = \frac{u_i}{w_i} = \frac{\mathbf{A}^\top \mathbf{X}_i}{\mathbf{C}^\top \mathbf{X}_i} \quad y_i = \frac{v_i}{w_i} = \frac{\mathbf{B}^\top \mathbf{X}_i}{\mathbf{C}^\top \mathbf{X}_i}$$

Leads to a system of equations, which is **linear in the parameters A, B and C**

## Estimating the Elements of P

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Rewrite

$$\begin{aligned} -\mathbf{X}_i^T \mathbf{A} &+ x_i \mathbf{X}_i^T \mathbf{C} = 0 \\ -\mathbf{X}_i^T \mathbf{B} &+ y_i \mathbf{X}_i^T \mathbf{C} = 0 \end{aligned}$$

as

$$\begin{aligned} \mathbf{a}_{x_i}^T \mathbf{p} &= 0 \\ \mathbf{a}_{y_i}^T \mathbf{p} &= 0 \end{aligned}$$

with

$$\begin{aligned} \mathbf{p} &= (p_k) = \text{vec}(\mathbf{P}^T) \\ \mathbf{a}_{x_i}^T &= (-\mathbf{X}_i^T, \mathbf{0}^T, x_i \mathbf{X}_i^T) \\ &= (-X_i, -Y_i, -Z_i, -1, 0, 0, 0, 0, x_i X_i, x_i Y_i, x_i Z_i, x_i) \\ \mathbf{a}_{y_i}^T &= (\mathbf{0}^T, -\mathbf{X}_i^T, y_i \mathbf{X}_i^T) \\ &= (0, 0, 0, 0, -X_i, -Y_i, -Z_i, -1, y_i X_i, y_i Y_i, y_i Z_i, y_i) \end{aligned}$$

$$\begin{aligned} \mathbf{a}_{x_i}^T \mathbf{p} &= (-X_i, -Y_i, -Z_i, -1, 0, 0, 0, 0, x_i X_i, x_i Y_i, x_i Z_i, x_i) \begin{bmatrix} p_{11} \\ p_{12} \\ p_{13} \\ p_{14} \\ p_{21} \\ p_{22} \\ p_{23} \\ p_{24} \\ p_{31} \\ p_{32} \\ p_{33} \\ p_{34} \end{bmatrix} \\ &= \left( \begin{array}{ccc} -\mathbf{X}_i^T, & \mathbf{0}, & x_i \mathbf{X}_i^T \end{array} \right) \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \\ \mathbf{C} \end{bmatrix} \\ &= -\mathbf{X}_i^T \mathbf{A} + x_i \mathbf{X}_i^T \mathbf{C} \end{aligned}$$



While it still holds:

$$\begin{bmatrix} \mathbf{a}_{x_1}^T \\ \mathbf{a}_{y_1}^T \\ \dots \\ \mathbf{a}_{x_i}^T \\ \mathbf{a}_{y_i}^T \\ \dots \\ \mathbf{a}_{x_I}^T \\ \mathbf{a}_{y_I}^T \end{bmatrix} \mathbf{p} = \underset{2I \times 12}{M} \underset{12 \times 1}{\mathbf{p}} \stackrel{!}{=} 0$$

## Solving Linear System $Ax = 0$

- In this form equivalent to finding null space of A
- Homogeneous System  $\rightarrow$  SVD
- Choose  $\mathbf{p}$  as the singular vector belonging to the singular value of 0

## Redundant Observations

$$M\mathbf{p} = \mathbf{w} \neq 0$$

- Find  $\mathbf{p}$  such that is minimizes

$$\begin{aligned} \Omega &= \mathbf{w}^T \mathbf{w} \\ \rightarrow \hat{\mathbf{p}} &= \arg \min_{\mathbf{p}} \mathbf{w}^T \mathbf{w} \\ &= \arg \min_{\mathbf{p}} \mathbf{p}^T M^T M \mathbf{p} \end{aligned}$$

With the constraint  $\|\mathbf{p}\|_2 = \sum_{ij} p_{ij}^2 = \|\mathbf{p}\| = 1$

- Singular value decomposition (SVD)

$$\underset{2I \times 12}{M} = \underset{2I \times 12}{U} \underset{12 \times 12}{S} \underset{12 \times 12}{V}^T = \sum_{i=1}^{12} s_i \mathbf{u}_i \mathbf{v}_i^T$$

- Choosing  $\mathbf{p} = \mathbf{v}_{12}$  (singular vector belonging to smallest singular value  $s_{12}$ ) minimises  $\Omega$

$$\mathbf{p} = \begin{bmatrix} p_{11} \\ \vdots \\ p_{34} \end{bmatrix} \rightarrow \mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}$$

## Critical Surfaces

- M of rank 11, if
  - Number of points  $\geq 6$
  - Assumption: No gross error

- **NO solution**, if all points  $\mathbf{X}_i$  are located on a **plane**

$$M = \begin{bmatrix} \vdots \\ \mathbf{a}_{x_i}^\top \\ \mathbf{a}_{y_i}^\top \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} -X_i & -Y_i & -Z_i & -1 & 0 & 0 & 0 & 0 & x_i X_i & x_i Y_i & x_i Z_i & x_i \\ 0 & 0 & 0 & 0 & -X_i & -Y_i & -Z_i & -1 & y_i X_i & y_i Y_i & y_i Z_i & y_i \\ \vdots & & & & & & & & & & & \end{bmatrix}$$

**e.g., assume all  $Z_i=0$**

$$M = \begin{bmatrix} \vdots \\ \mathbf{a}_{x_i}^\top \\ \mathbf{a}_{y_i}^\top \\ \vdots \end{bmatrix}$$

$$= \begin{bmatrix} -X_i & -Y_i & 0 & -1 & 0 & 0 & 0 & 0 & x_i X_i & x_i Y_i & 0 & x_i \\ 0 & 0 & 0 & 0 & -X_i & -Y_i & 0 & -1 & y_i X_i & y_i Y_i & 0 & y_i \\ \vdots & & & & & & & & & & & \end{bmatrix}$$

**rank deficiency**

- (All points  $\mathbf{X}_i$  and the projection center  $\mathbf{X}_O$  are located on a twisted cubic curve)

**$P$  from  $K, R, \mathbf{X}_O$**

$$P = [KR] - KR\mathbf{X}_O = [H|\mathbf{h}]$$

- Projection Center:

$$\mathbf{X}_O = -H^{-1}\mathbf{h}$$

- Rotation matrix:

- QR decomposition of  $H^{-1}$

$$H^{-1} = (KR)^{-1} = R^{-1}K^{-1} = R^T K^{-1} = QR$$

- Calibration Matrix:

- normalise due to homogeneity of H (-> homogeneity of K)

$$K \leftarrow \frac{1}{K_{33}} K$$

- K has **positive** diagonal elements
- For **negative camera constant**, choose

$$K \leftarrow KR(z, \pi) \quad R \leftarrow R(z, \pi)R$$

- using:

$$R(z, \pi) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

## DLT in a Nutshell

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### 1. Build the M for the linear system

$$M = \begin{bmatrix} \mathbf{a}_{x_1}^\top \\ \mathbf{a}_{y_1}^\top \\ \vdots \\ \mathbf{a}_{x_I}^\top \\ \mathbf{a}_{y_I}^\top \end{bmatrix} \quad \begin{array}{c} M \mathbf{p} \stackrel{!}{=} 0 \\ \uparrow \quad \uparrow \\ 2I \times 12 \quad 12 \times 1 \end{array}$$

with

$$\begin{aligned} \mathbf{a}_{x_i}^\top &= (-X_i, -Y_i, -Z_i, -1, 0, 0, 0, 0, x_i X_i, x_i Y_i, x_i Z_i, x_i) \\ \mathbf{a}_{y_i}^\top &= (0, 0, 0, 0, -X_i, -Y_i, -Z_i, -1, y_i X_i, y_i Y_i, y_i Z_i, y_i) \end{aligned}$$

## DLT in a Nutshell

### 2. Solve by SVD $M = U S V^\top$ Solution is last column of $V$

$$\mathbf{p} = \mathbf{v}_{12} \Rightarrow P = \begin{bmatrix} p_1 & p_2 & p_3 & p_4 \\ p_5 & p_6 & p_7 & p_8 \\ p_9 & p_{10} & p_{11} & p_{12} \end{bmatrix}$$

# DLT in a Nutshell

## 3. If individual parameters are needed

$$P = KR [I_3 | -X_O] = [H|h]$$

$$X_O = -H^{-1} h$$

$$QR(H^{-1}) = R^T K^{-1}$$

$$R \leftarrow R(z, \pi) R$$

$$K \leftarrow \frac{1}{K_{33}} KR(z, \pi)$$

- Solution is not statistically optimal
- Direct solution, no initial guess needed